

# Atrium Radiant Heating System Atascadero, CA

Isaac Golan & Julian Rubinstein

March 13, 2025

# Problem Statement

- A new community center is being built in Atascadero, California.
- The owner wants to install a radiant heating system in the atrium.
- The radiant heating system will pump heated water in tubes embedded in the concrete foundation.
- As a result, the concrete will heat up and transfer heat to the atrium air space
- Desired indoor air temperature: 68 °F

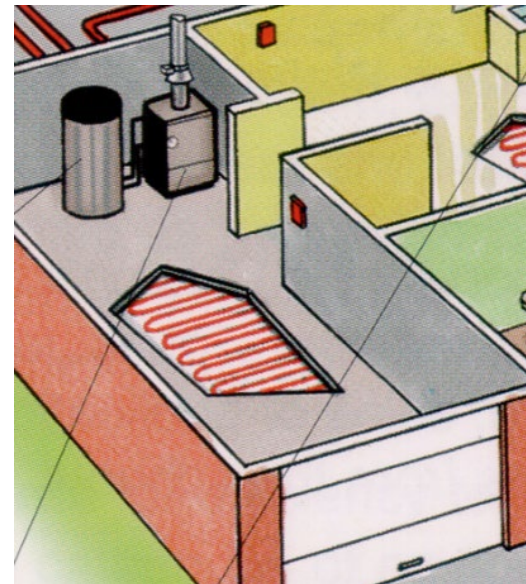
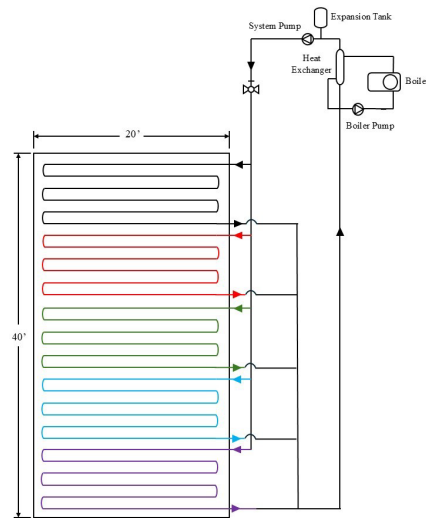


Table 1. Temperature Data for Atascadero, CA

| Month | Mean<br>Daily Max | Mean<br>Daily Min | Record<br>High | Record<br>Low |
|-------|-------------------|-------------------|----------------|---------------|
| Jan   | 60.7              | 32.8              | 79             | 11            |
| Feb   | 65                | 36                | 83             | 18            |
| Mar   | 68                | 37.2              | 91             | 20            |
| Apr   | 73.6              | 39.5              | 98             | 25            |
| May   | 79.9              | 43.2              | 103            | 30            |
| Jun   | 87                | 47.3              | 111            | 31            |
| Jul   | 93.9              | 50.1              | 113            | 37            |
| Aug   | 93.6              | 49.6              | 108            | 37            |
| Sep   | 89.9              | 46.5              | 111            | 33            |
| Oct   | 81.6              | 40.9              | 105            | 19            |
| Nov   | 69.2              | 35.4              | 95             | 14            |
| Dec   | 61.7              | 31.9              | 87             | 15            |

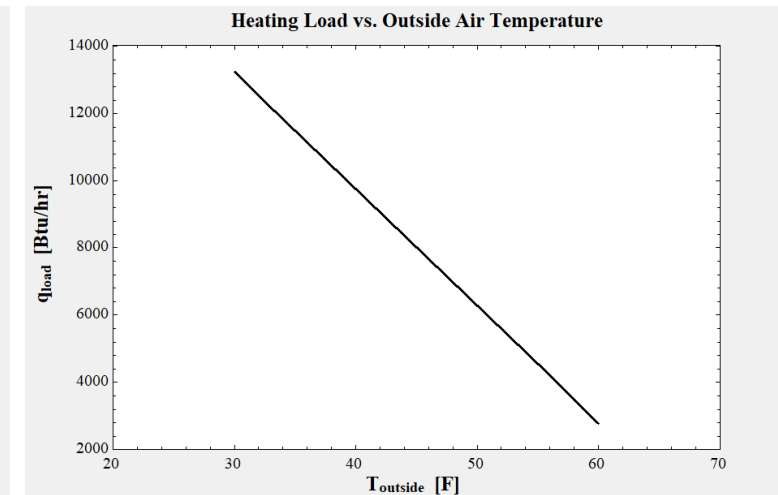
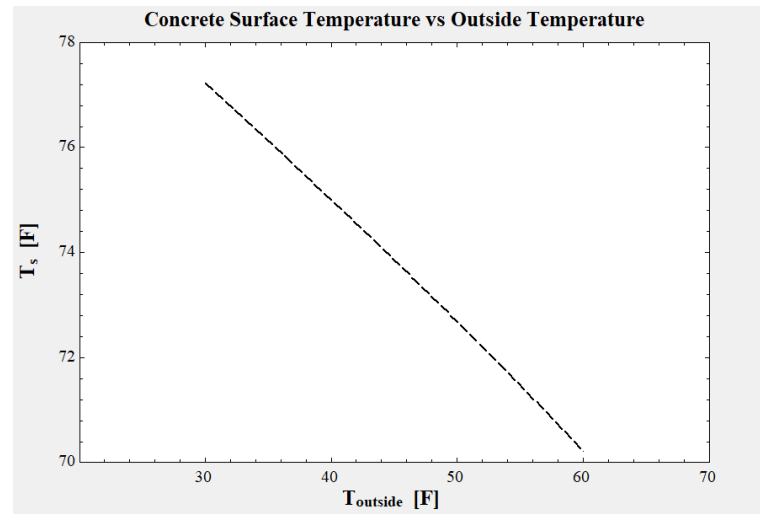
# Overview

|                              |                      |
|------------------------------|----------------------|
| Design Day Temperature       | 30°F                 |
| Indoor Set Point Temperature | 68°F                 |
| Window Height                | 30 ft                |
| Window Area                  | 1200 ft <sup>2</sup> |
| Concrete Surface Area        | 800 ft <sup>2</sup>  |
| Emissivity of All Surfaces   | 0.9                  |



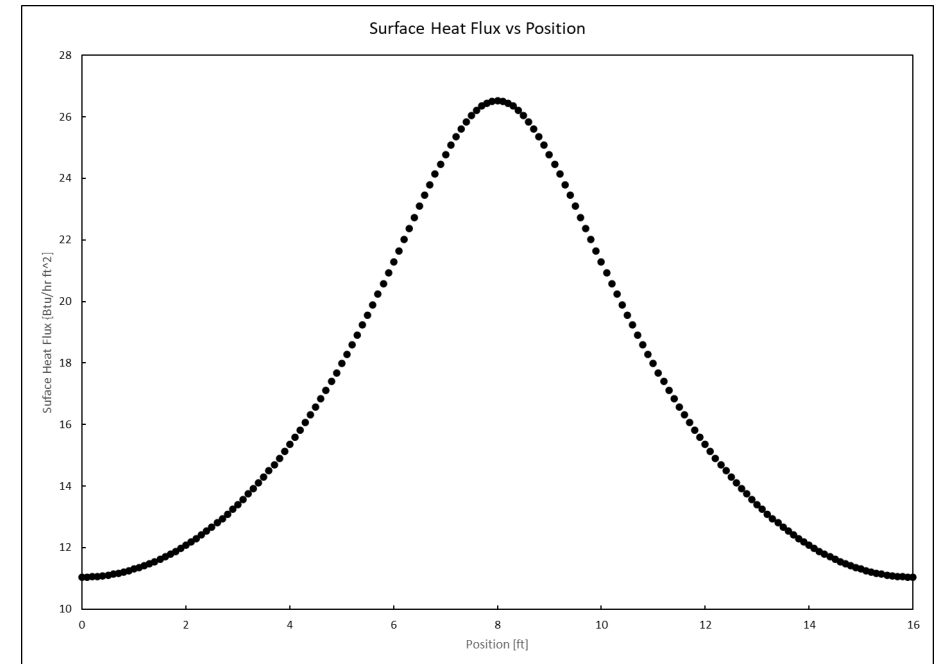
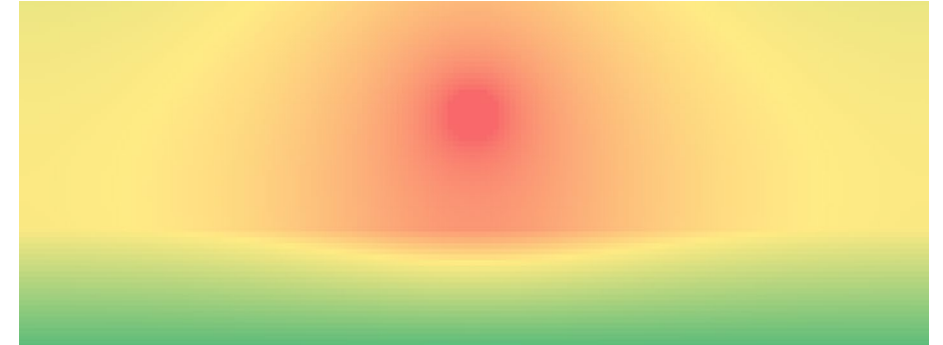
- Conduction through window
- Natural convection
- Radiation exchange between concrete and window

- PEX Tubing
- Heating Load:
  - **15,738 Btu/hr**
- Startup time:
  - **6 hours from 55 °F**
- Overall cost:
  - **\$19,204.38**



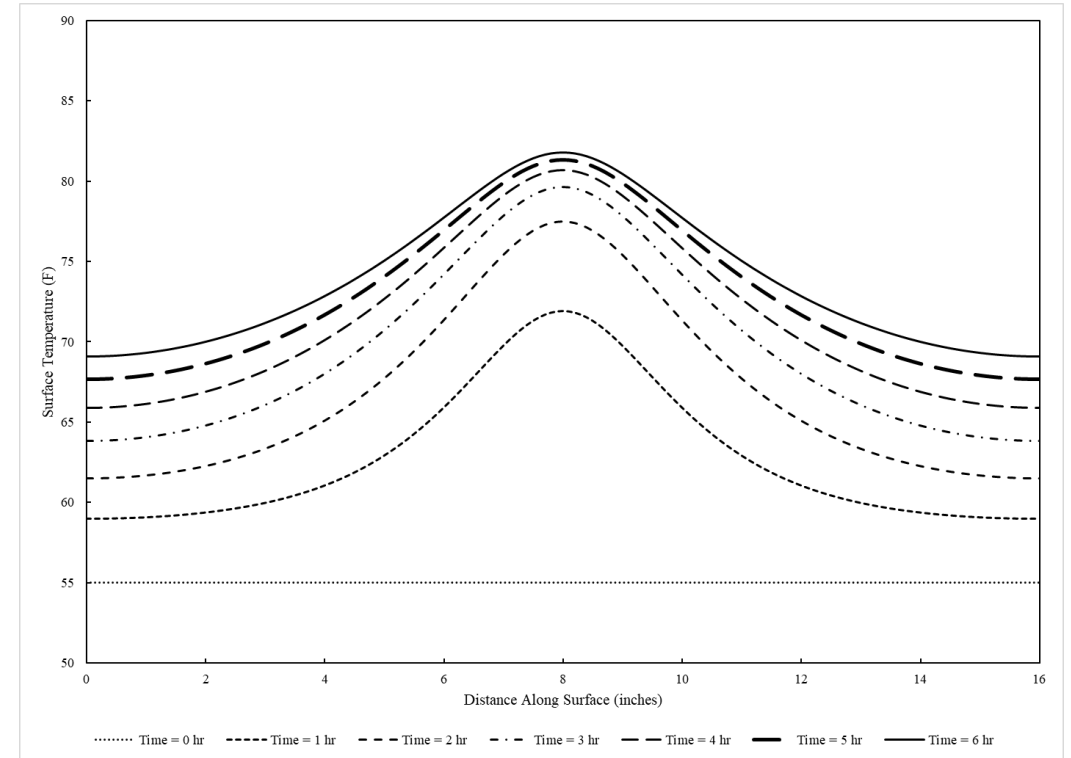
# Steady State Model

- Required surface heat load:  
13224 Btu/hr
- Resultant surface heat load:  
13495 Btu/hr
- Pipe surface temperature:  
72 °F



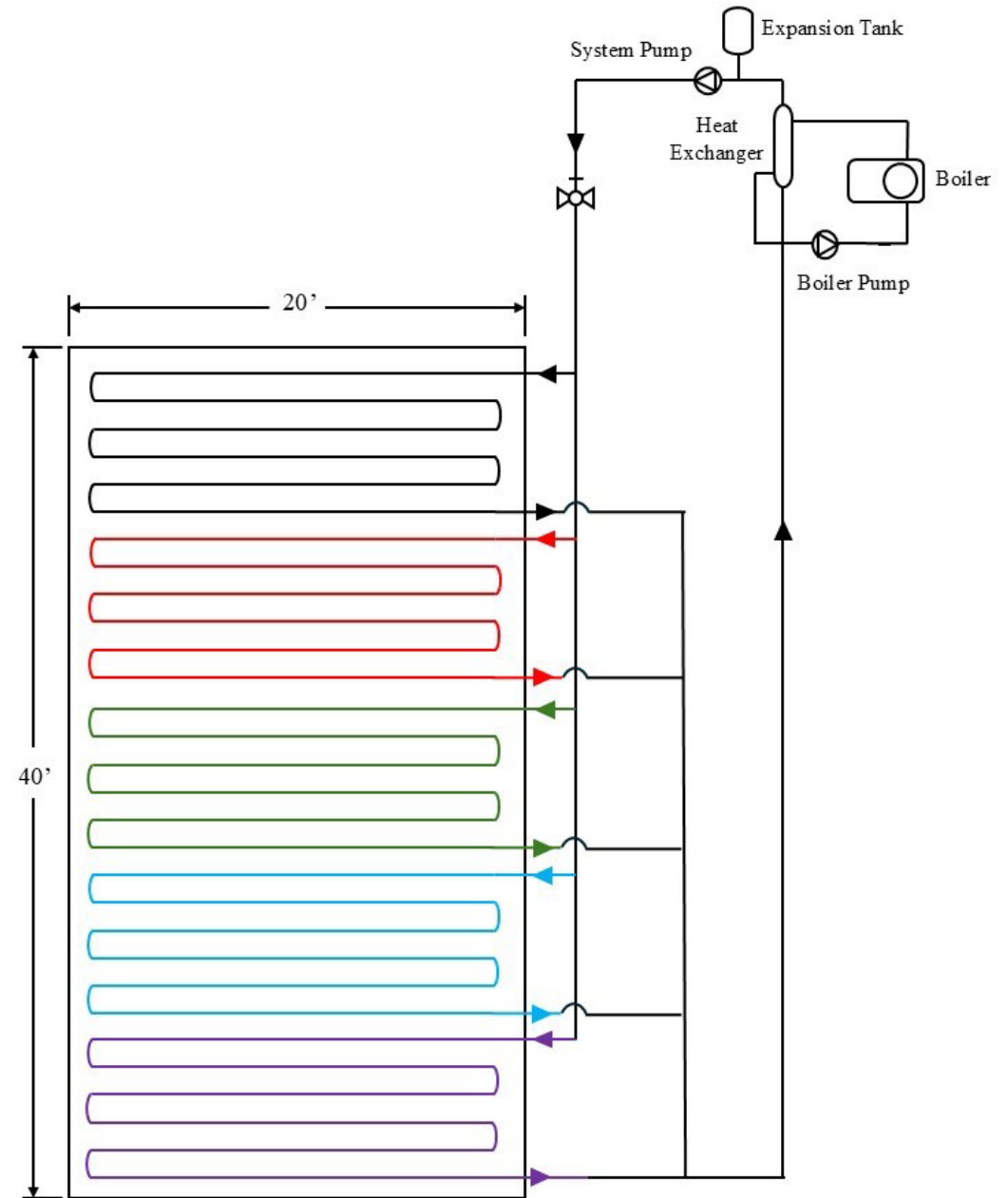
# Transient Model

- Startup time (from 55 °F):  
6 hours
- Temperature after cooldown:  
67.8 °F
- Reheat time (from 67.8 °F):  
3 hours

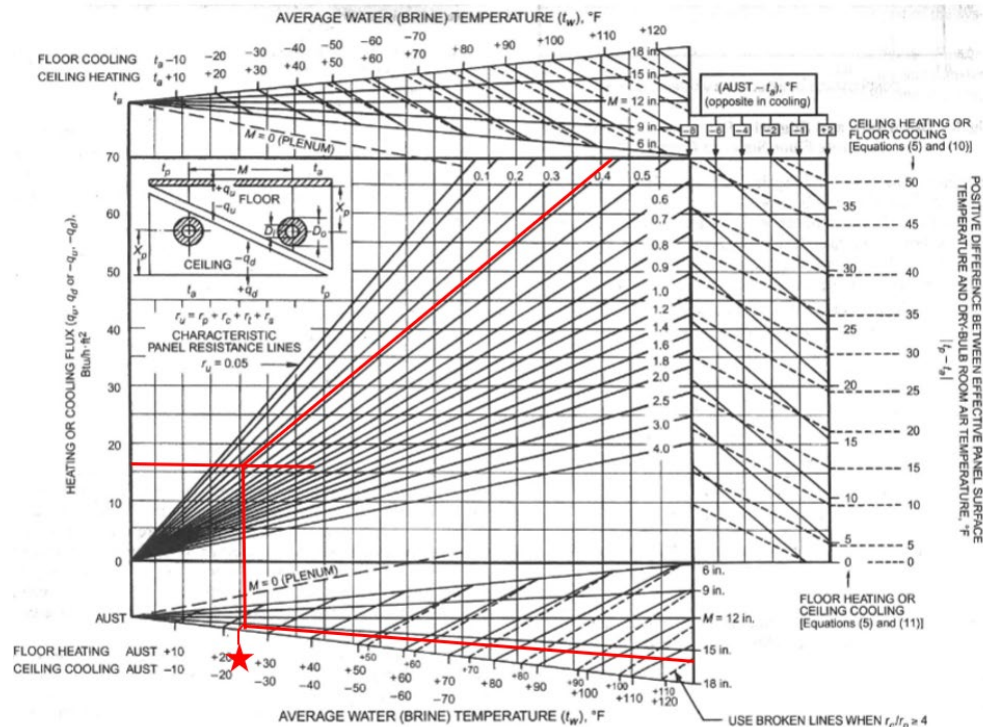


# Piping

|                                    |                          |
|------------------------------------|--------------------------|
| Length of Each Tube                | 120 ft                   |
| Total Number of Loops              | 5                        |
| Flowrate Through Each Loop         | 2085 lbm/hr {5.043 gpm}  |
| Temperature Drop Through Each Loop | 1.3 °F                   |
| Required Loop Inlet Temperature    | 93.73 °F                 |
| Total Heat Transfer Rate           | 15,738 Btu/hr            |
| Final Pressure Drop in Each Loop   | 4.64 psi {10.76 ft head} |

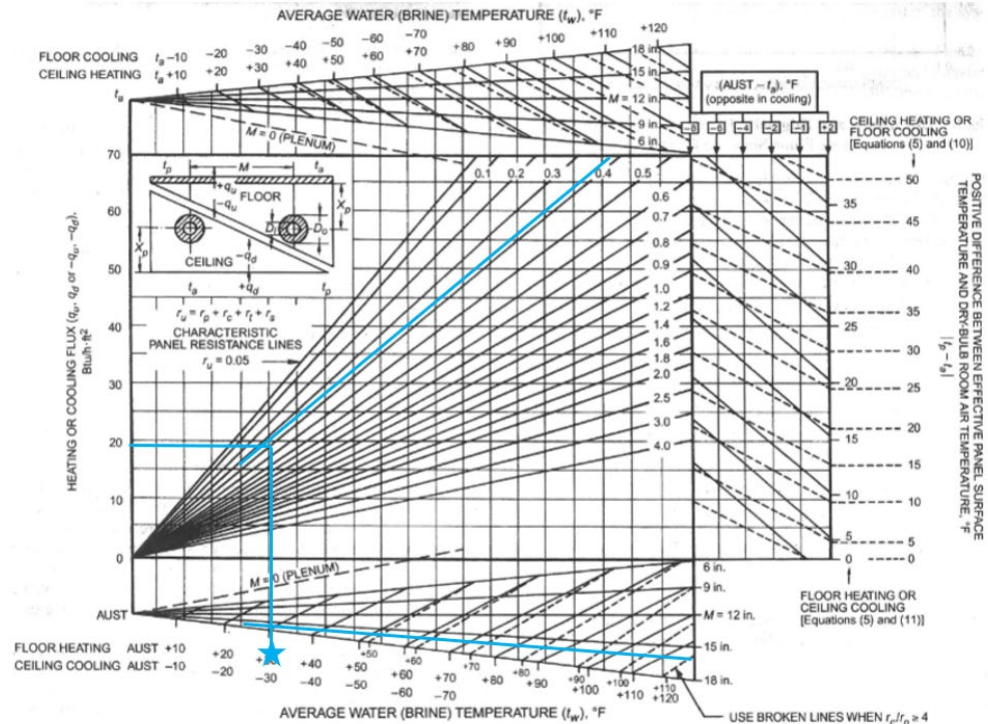


# ASHRAE Method



Using **surface heat flux (13224 Btu/hr)**

- $23^{\circ}\text{F} + 68^{\circ}\text{F} = 91^{\circ}\text{F}$
- $\sim 5^{\circ}\text{F}$  underestimation from  $T_m = 96.26^{\circ}\text{F}$

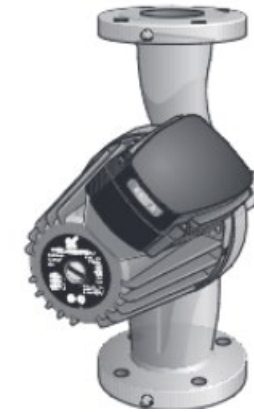
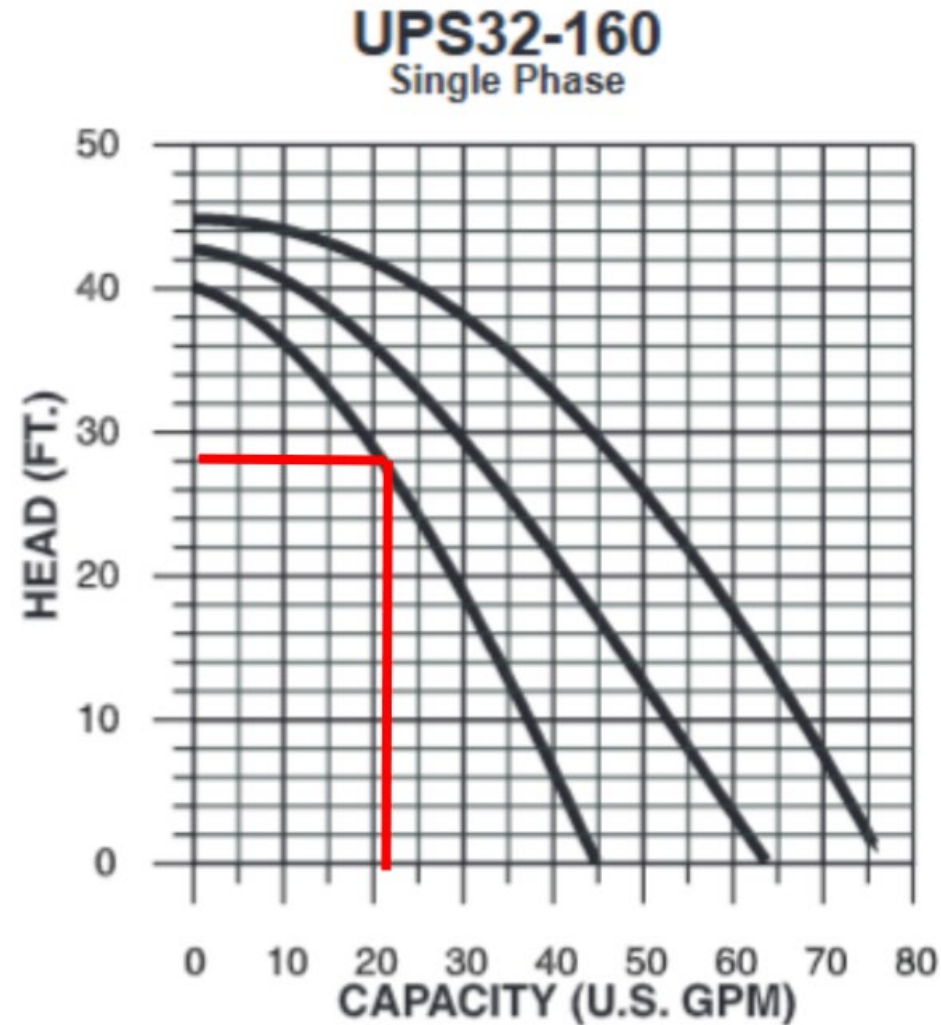


Using **total heat flux (15,738 Btu/hr)**

- $30^{\circ}\text{F} + 68^{\circ}\text{F} = 98^{\circ}\text{F}$
- $\sim 2^{\circ}\text{F}$  overestimation from  $T_m = 96.26^{\circ}\text{F}$

# System Pressure Drop and Pump Selection

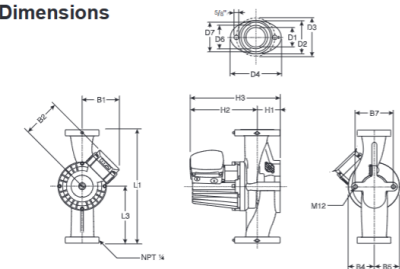
- **VersaFlo**  
**UPS32-160**  
**Single Phase**
- Head:
  - **28 ft**
- Flowrate
  - **21 gpm**
- Mechanical Power
  - **0.2481 hp**
- Efficiency
  - **60%**



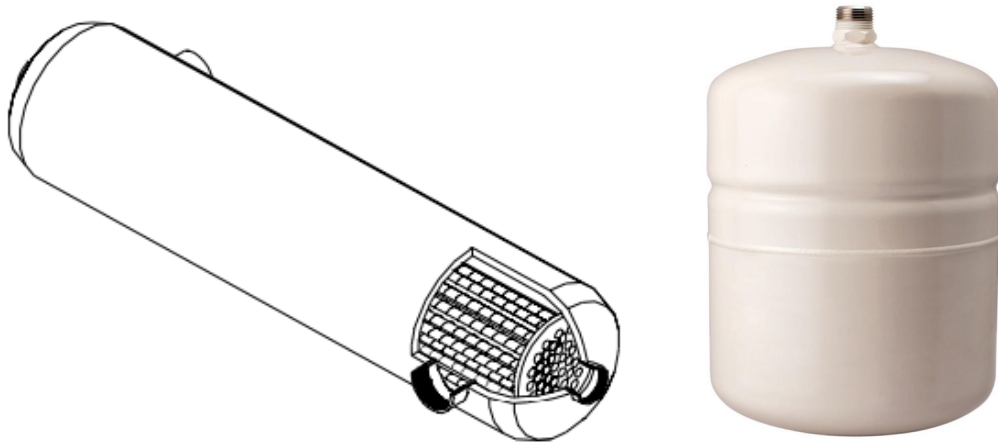
## Technical Data

FLOW RANGE: 9 – 78 U.S. GPM  
HEAD RANGE: 1 – 46 FT.  
MAXIMUM WORKING PRESSURE: 145 PSI  
FLANGES: 1 1/4" and 1 1/2", 2-Bolt with (2) 5/8" diameter holes  
MINIMUM FLUID TEMPERATURE: 14°F (-10°C)  
MAXIMUM FLUID TEMPERATURE (Open Systems): 140°F\* (60°C)  
MAXIMUM FLUID TEMPERATURE (Closed Systems): 230°F\* (110°C)  
AMBIENT AIR TEMPERATURE: 32°F (0°C) to 104°F (40°C)  
\*MAXIMUM TEMPERATURE: 220°F, 104°C. NOTE: It is recommended to keep the operating temperature as low as possible (ie. below 140°F, 60°C to avoid precipitation of calcium)

## Dimensions



# Heat Exchanger & Expansion Tank



- **SEC PL45 Shell & Tube HX**
  - $UA = 454.4 \text{ Btu/hr-}^{\circ}\text{F}$
  - 5.5% boiler flowrate to achieve heat load
  - **correction factor of 0.9716.**
- **Watts 2.1 Gallon Water Expansion Tank, sized for:**
  - Specific volume at Min and Max temp
  - Minimum pressure above NPSHR = 14.7 psia
  - Maximum system pressure = 60 psia

| Nominal Capacity |        | Hot water |     |               |      | Cold Water |      |               |      | Heat Transfer Surface |      |
|------------------|--------|-----------|-----|---------------|------|------------|------|---------------|------|-----------------------|------|
|                  |        | Flow      |     | Pressure Drop |      | Flow       |      | Pressure Drop |      |                       |      |
| kW               | Btu/hr | l/min     | gpm | kPa           | psig | l/min      | gpm  | kPa           | psig | m²                    | ft²  |
| 13               | 45,000 | 23        | 6.1 | 6.2           | 0.90 | 150        | 39.6 | 7.4           | 1.07 | 0.150                 | 1.62 |

# Cost Estimation

- Total upfront cost: \$16397.44
- Yearly energy cost: \$289.01
- Total present value (15-year life): \$19204.38
- Electric system present value: \$31772.80



# Controls and Next Steps

- Thermostat control
- Control valves in the heating loop
- Controlling the boiler to further reduce energy consumption.
- This will require upfront costs for materials, installation, and programming, as well as future maintenance and electricity costs.